

Dice Probability

Time limit: 1.00 s

Memory limit: 512 MB

You throw a dice n times, and every throw produces an outcome between 1 and 6. What is the probability that the sum of outcomes is between a and b ?

Input

The only input line contains three integers n , a and b .

Output

Print the probability rounded to six decimal places (rounding half to even).

Constraints

- $1 \leq n \leq 100$
- $1 \leq a \leq b \leq 6n$

Example

Input	Output
2 9 10	0.194444